

# Specific decomposition rate constants in unimolecular reactions<sup>1</sup>

HUW O. PRITCHARD AND GALINA M. DIKER

*Centre for Research in Experimental Space Science, York University, Downsview, Ont., Canada M3J 1P3*

AND

ANDREW W. YAU

*Herzberg Institute of Astrophysics, National Research Council of Canada, Ottawa, Ont., Canada K1A 0R6*

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The original treatment of unimolecular reactions given by Polanyi and Wigner in 1928 is recast in quantum terms. It is found that very simple and natural choices for the oscillation which is assumed to cause reaction lead to good predictions of the specific reaction rate constant  $k(E)$ , and also the overall rate of reaction: nitrous oxide, carbon dioxide, methyl isocyanide, trideuteriomethyl isocyanide, ethyl isocyanide, ethyl chloride, cyclopropane, cyclobutane, and ethane are treated as illustrations.

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Le travail original de Polanyi et Wigner (1928) sur les réactions unimoléculaires est revu en termes quantiques. Un choix naturel et simple de l'oscillation qui cause présumément la réaction fournit de bonnes prédictions pour les constantes de vitesse spécifiques de réaction  $k(E)$  aussi bien que la vitesse globale de réaction. Comme exemples, on utilise l'oxide nitreux, l'oxide carbonique, l'isocyanure de méthyle, l'isocyanure de tridéutériométhyle, l'isocyanure d'éthyle, le chlorure d'éthyle, le cyclopropane, le cyclobutane et l'éthane.

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## Introduction

In 1928, Polanyi and Wigner proposed the first molecular-dynamic picture of a unimolecular reaction (1). They imagined the molecule to consist of  $n$  atoms of essentially equal mass, held together by roughly equally strong forces; they then analysed in simple terms the vibrational motions of such a collection of atoms and concluded that an amount of energy greater than  $\epsilon$  would be concentrated in a particular bond approximately  $\nu e^{-\epsilon/kT}$  times per second, where  $\nu$  is the fastest vibrational frequency in the molecule, and  $\epsilon \gg kT$ . Thus, if  $\epsilon$  is identified as the critical energy required for dissociation or for transformation to another species (we shall use  $E_{\text{crit}}$  for this quantity in our reformulation), then it follows naturally that the rate constant for the reaction is, in its simplest form,

$$[1] \quad k \approx \nu e^{-\epsilon/kT}$$

This expression applies only if the critical energy  $\epsilon$  is assumed to be confined in one linear dimension, but Polanyi and Wigner also derived formulae applicable to the cases where  $\epsilon$  was assumed to be concentrated in either two or three linear dimensions, the latter yielding the rate constant expression

$$[2] \quad k \approx \frac{2\nu\epsilon}{kT} e^{-\epsilon/kT}$$

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This primitive treatment of molecular vibration was developed extensively by Slater (2) in a slightly different form, but nevertheless one which was foreseen by Polanyi and Wigner in their original discussion: he defined a critical coordinate, and attempted to calculate the mean frequency with which, in an oscillating molecule, this coordinate would exceed a certain length, and reaction was assumed to have taken place at this point. Using a harmonic-oscillator model for cyclopropane, Slater calculated the shape of the fall-off curve for the isomerisation to propylene (3) simultaneously with and independently of (ref. 2, p. 181) our measurements (5) on this reaction, and his theory was received with tremendous enthusiasm as the result of the remarkable agreement between his predictions and the experiment. Later, however, as Slater tried to extend his approach to anharmonic oscillators and to quantised oscillators (ref. 2, Chapt. 10), and because of insuperable difficulties relating to degeneracy changes upon partial isotopic substitution of highly symmetrical molecules, e.g., *cyclo*-C<sub>3</sub>H<sub>6</sub> and *cyclo*-C<sub>3</sub>H<sub>4</sub>D<sub>2</sub>, or CH<sub>3</sub>NC and CH<sub>2</sub>DNC (7), the theory has fallen into disrepute. This, taken together with other unsuccessful attempts based upon eq. [2] to account for high frequency factors (8, 9), seems an unfitting fate for such an attractive concept.

In this paper, we present a quantum formulation for the approximate rate of decomposition of a vibrationally excited molecule, based upon the origi-

nal concept of Polanyi and Wigner (1), and we show, within the limitations of the available spectroscopic data and our understanding of the nature of molecular vibrations, that acceptable results can be obtained.

### Basic Formulation

We begin by regarding the molecule as a collection of  $s$  independent oscillators and rigid rotors, with a set of vibration-rotation states  $i = \{n_1, n_2, n_3, \dots, n_s\}$  at energies

$$[3] \quad E_i = \sum_{j=1}^s \epsilon_j(n_j)$$

where  $\epsilon_j(n_j)$  is the energy associated with the  $j$ th oscillator/rotor in the quantum state  $n_j$ . We then attempt to represent the reaction process by the motion of a particular oscillator,  $r$ , and assume that reaction can occur if more energy than the critical energy of the reaction,  $E_{\text{crit}}$ , is found in this oscillator, i.e.

$$[4] \quad \epsilon_r \geq E_{\text{crit}}$$

Furthermore, we assume, in the spirit of Polanyi and Wigner (1), that if a molecule finds itself in a state in which it has more energy than  $E_{\text{crit}}$  in oscillator  $r$ , it will decompose within the vibrational period of that oscillator: for either a harmonic or a Morse oscillator, this is

$$[5] \quad \tau = (2c\nu_r)^{-1} \text{ seconds}$$

where  $c$  is the velocity of light and  $\nu_r$  is the vibrational frequency of the oscillator  $r$  in wavenumber units.

If we now denote the state density of the molecule at energy  $E$  by  $\rho_s(E)$  and the corresponding state density of a *hypothetical submolecule* with oscillator  $r$  removed by  $\rho_{s-1}^r(E)$ , then the fraction of states at  $E$  that decompose to products with time constant  $\tau^{-1}$  is

$$[6a] \quad \alpha_1(E) = \sum \rho_{s-1}^r [E - \epsilon_r(n_r)] / \rho_s(E)$$

where the summation is over  $n_r$  for which  $E_{\text{crit}} \leq \epsilon_r(n_r) \leq E$ ; in the quantity  $\rho_{s-1}^r(E)$ , the subscript  $s-1$  refers to the remaining  $(s-1)$  oscillators in the hypothetical submolecule when the oscillator  $r$  has been removed. The specific rate constant is then

$$[7a] \quad k_1(E) = 2c\nu_r \alpha_1(E)$$

Equations [6a] and [7a] are based on the postulate that the critical energy is localised in one oscillator, and are thus the quantum analogue of eq. [1]. In a more complex molecule where we may envisage the critical energy to be shared between  $m$

of the  $s$  oscillators, eq. [6a] is generalised to

$$[6b] \quad \alpha_m(E) = \sum \rho_{s-m}^{r_1 r_2 \dots r_m} (E - \epsilon_R) g_m(\epsilon_R) / \rho_s(E)$$

where  $\rho_{s-m}^{r_1 r_2 \dots r_m}(E)$  is the state density of the hypothetical submolecule with the oscillators  $r_1, r_2, \dots, r_m$  removed,

$$\epsilon_R = \epsilon_{r_1}(n_{r_1}) + \epsilon_{r_2}(n_{r_2}) + \dots + \epsilon_{r_m}(n_{r_m})$$

is the energy in the reactive oscillators  $r_1, r_2, \dots, r_m$  in the state  $\{n_{r_1}, n_{r_2}, \dots, n_{r_m}\}$ ,  $g_m(\epsilon_R)$  is the state degeneracy of the group of  $m$  interesting oscillators at energy  $\epsilon_R$ , and the summation is over all values of  $n_r$  for which  $E_{\text{crit}} \leq \epsilon_R \leq E$ . The average lifetime of these states is, following Polanyi and Wigner (1), that of the oscillator with the highest frequency, viz.  $|\nu_r|_{\text{max}}$ , whence [7a] is generalised to

$$[7b] \quad k_m(E) = 2c|\nu_r|_{\text{max}} \alpha_m(E)$$

The computation of  $k(E)$  using eqs. [6] and [7] requires as input parameters the relevant vibration-rotation constants of the molecule and the critical energy of the reaction. Since, except for the simplest of molecules, we have no *a priori* knowledge of  $E_{\text{crit}}$ , we will use the observed infinite-pressure activation energy,  $E_\infty$ , as the critical energy,  $E_{\text{crit}}$ . In addition, for the sake of logical consistency, we adjust the chosen value of  $\nu_r$  (or else  $|\nu_r|_{\text{min}}$ , the smallest of the group of  $m$  frequencies) very slightly to ensure that there is a reactive level *exactly* at  $E_{\text{crit}}$ ; typically, this adjustment amounts only to a few percent of the value of  $\nu_r$  (or  $|\nu_r|_{\text{min}}$ ). We will proceed to examine the behaviour of eqs. [6] and [7] in considerable detail for two test cases, the thermal isomerisations of methyl isocyanide and of cyclopropane, and thereafter will present more briefly applications to other reactions, and discussions of various points which arise in respect of the formulation and of the results.

### Some Preliminaries

In the discussion which follows, we will always compare our synthetic  $k(E)$  function with those derived for the same molecular model using the Slater-Forst and RRKM procedures; moreover, with a single exception noted below, we will always regard the molecule as a collection of rigid rotating Morse oscillators. Thus, before tests of eqs. [6] and [7] can be made, it is necessary to describe the nature of such tests.

#### (i) Slater-Forst Procedure

We take as the target against which to compare our synthetic  $k(E)$  function, the empirical specific rate function  $k(E)$  derived from the experimental

rate law by the Slater-Forst procedure (10, 11), i.e.

$$[8] \quad \bar{k}(E) = A_{\infty} \rho(E - E_{\infty}) / \rho(E)$$

where  $E_{\infty}$  and  $A_{\infty}$  are respectively the infinite-pressure activation energy and frequency factor: we have given detailed arguments (12) why [8] should provide a reasonable estimate for the specific rate function, and more recently, we have made detailed numerical tests of its accuracy (13).

The specific rate function  $\bar{k}(E)$  given by eq. [8] is not entirely unique, but depends slightly upon the assumed molecular model, as is shown in Fig. 1 for methyl isocyanide. Of the three curves shown, the topmost one assumes that the molecular oscillations are harmonic, whereas the bottom one assumes that they are Morse oscillators with the anharmonicity factors we estimated previously (12): we have concluded since that those empirical anharmonicity factors were much too severe, and we have evolved a new empirical scheme (which is described in the Appendix), based upon the assumed dissociation behaviour of the oscillator in question; this scheme will be used throughout this paper, and the resulting  $\bar{k}(E)$  function for methyl isocyanide lies between the other two curves.

#### (ii) RRKM Calculations

The RRKM specific rate functions  $k^{\ddagger}(E)$  shown for comparison in the following sections are com-

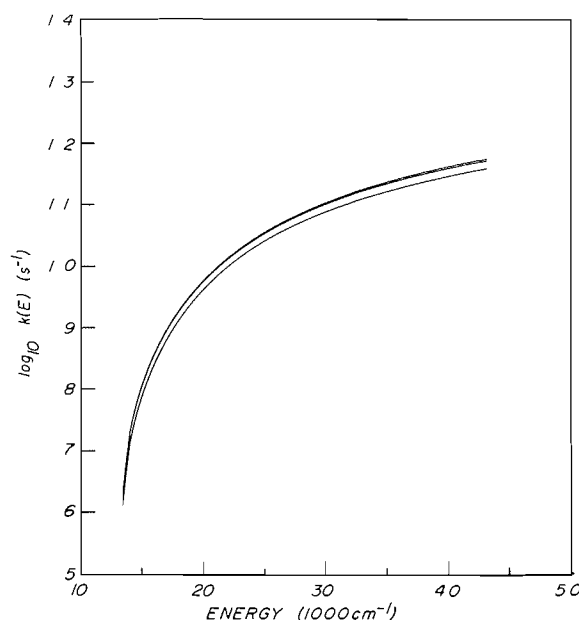


FIG. 1. Comparison of three empirical  $\bar{k}(E)$  functions for methyl isocyanide, which are, in order: topmost curve, harmonic oscillators; middle curve, Morse oscillators with anharmonicity factors defined by eq. [A1]; bottom curve, Morse oscillators with anharmonicity factors from ref. 12.

puted by a relatively simple procedure, roughly equivalent in its degree of sophistication to our model described by eqs. [6] and [7]:  $k^{\ddagger}(E)$  is calculated using eq. [6.20] of ref. 7, assuming no adiabatic rotations; the program used was essentially that supplied by Holbrook (ref. 7, p. 149), but modified for Morse oscillators, and with the sums and densities of states calculated by our own techniques (15). The calculations of  $k(E)$ ,  $\bar{k}(E)$ , and  $k^{\ddagger}(E)$  are always mutually consistent in the sense that they use the same set of frequencies, anharmonicities, and rotational constants for the molecule in question; likewise, all calculations use the same value of  $E_{\infty}$ , and the same value of  $A_{\infty}$  is used in the Slater-Forst procedure and in the RRKM frequency adjustment. In the latter case, after the choice of the reaction coordinate (which is usually taken to be one of the frequencies chosen as  $\nu_r$ ), the two lowest frequencies remaining in the molecule were always adjusted to achieve the matching with the entropy of activation: the results for  $k^{\ddagger}(E)$ , of course, are insensitive to the choice of reaction coordinate (cf. Fig. 3 below) provided the assumed activated complex has the required entropy.

#### (iii) The Overall Rate Constant

As is shown in Fig. 1, the numerical magnitudes of  $\bar{k}(E)$ , and also  $k^{\ddagger}(E)$ , are model-dependent and as can be seen in subsequent diagrams,  $\bar{k}(E)$  and  $k^{\ddagger}(E)$  can often be markedly different although they always correspond to the same value of the rate constant at the RRKM fitting temperature. Thus, whilst our primary aim in this work has been to compare the synthetic  $k(E)$  function with the empirical  $\bar{k}(E)$  function, a complete test of the synthetic  $k(E)$  function requires that it also give acceptable values for the overall activation energy and frequency factor. Consequently, for each synthetic  $k(E)$  function, (and also for each empirical  $\bar{k}(E)$  function), we have always reconstructed the overall rate constant for the reaction using

$$[9] \quad k_{\infty}(T) = [Q(T)]^{-1} \int_0^{\infty} \rho(E) k(E) e^{-E/RT} dE$$

where  $Q(T)$  is the reactant partition function, and compared the synthetic overall rate constant with the observed value. We also examined the derived Arrhenius parameters  $A_{\infty}$  and  $E_{\infty}$  for constancy with temperature. In the case of the  $\bar{k}(E)$  function, this simply served as a check on the correctness and consistency of the partition functions and densities of states used in the calculations. On the other hand, using the synthetic  $k(E)$  functions there were small, and real, variations of up to 0.2 or 0.3 kcal mol<sup>-1</sup> in  $E_{\infty}$  and up to about  $\pm 30\%$  in  $A_{\infty}$  over

a temperature range of a factor of two in  $T$ . Consequently, to complete the test of each model calculation, the computed values of  $A_\infty$  and  $E_\infty$  in the appropriate experimental temperature range are listed in the caption of each diagram, along with the observed values, the latter in italics.

### Methyl Isocyanide

There have been several theoretical studies of the potential energy surface for the transition between methyl isocyanide and methyl cyanide (16–18), and it is clear that if one had to postulate a single vibrational mode which, if sufficiently excited, would lead to reaction, one would naturally choose the C—N—C bending mode  $\nu_8 = 263 \text{ cm}^{-1}$  (19). Figure 2 shows the result of the application of eqs. [6a] and [7a] for this model, with  $\nu_8 = 263 \text{ cm}^{-1}$  adjusted slightly to  $\nu_r = 267 \text{ cm}^{-1}$  to ensure that a reactive level occurs exactly at the value of the critical energy. Also shown on the same diagram are the RRKM  $k^*(E)$  function and the empirical  $\bar{k}(E)$  function, and the agreement with the latter is remarkable; likewise, the agreement of the overall rate constant with experiment, as documented in the legend to the diagram, is excellent.

#### Sensitivity of the Computed Results to the Nature of the Assumed Model

As we noted above, and showed in Fig. 1, the computed values of  $k(E)$  depend very slightly upon the oscillator model assumed for the reacting molecule, but nevertheless, any of these functions will reproduce the observed rate constant when substituted in eq. [9]; likewise, the  $k^*(E)$  function calculated by the RRKM prescription, although significantly different from  $k(E)$ , still corresponds to the same value of the rate constant at the matching temperature.<sup>2</sup> Inspection of eqs. [6a] and [7a], however, shows that our synthetic  $k(E)$ , and the derived overall rate constant, should be quite sen-

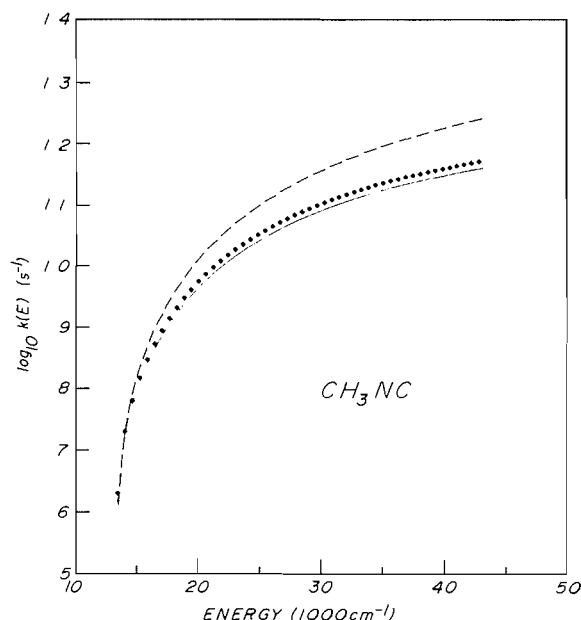


FIG. 2. Comparison of synthetic  $k(E)$ , empirical  $\bar{k}(E)$ , and RRKM  $k^*(E)$  functions for methyl isocyanide using anharmonicities defined by eq. [A1]: solid line,  $k(E)$  using  $\nu_r = 263 \text{ cm}^{-1}$ ; individual points,  $k(E)$ ; dashed line,  $k^*(E)$ . The rate constants calculated from  $k(E)$  using eq. [9] give the following Arrhenius parameters:  $E_\infty = 38.2$  (38.2)  $\text{kcal mol}^{-1}$ ,  $A_\infty = 1.9$  ( $2.2 \times 10^{13} \text{ s}^{-1}$ ) at 500 K.

sitive to the choice of the reactive oscillator  $\nu_r$ , and the nature of this sensitivity is shown in Fig. 3a. In this diagram, [6a] and [7a] were applied to the rotating harmonic-oscillator model of methyl isocyanide and  $\nu_r$  was considered to be, in turn,  $\nu_8 = 263$ ,  $\nu_4 = 945$ ,  $\nu_3 = 1429$ ,  $\nu_2 = 2166$ , and  $\nu_1 = 2966 \text{ cm}^{-1}$ . Most of the sensitivity of  $k(E)$  to  $\nu_r$  arises in eq. [7a] and, in fact, for harmonic oscillators  $k(E)$  is directly proportional (to within about 3 or 4%) to  $\nu_r$  at any value of  $E$ ; thus, the computed overall rate constant is also approximately proportional to  $\nu_r$ , with  $A_\infty \approx (1.9 \times 10^{13}) \times (\nu_r/267) \text{ s}^{-1}$  in this case; this proportionality does not hold, however, as we shall see below for strongly anharmonic oscillators.

In contrast, Fig. 3b shows the RRKM results for  $k^*(E)$  where, in turn, each of the five frequencies treated as  $\nu_r$  in Fig. 3a is regarded as the critical reaction coordinate; this figure again demonstrates the now well-known insensitivity of the RRKM  $k^*(E)$  function to the choice of reaction coordinate, which arises through the frequency adjustment procedure to match the experimental  $A$ -factor (12). Thus, we see that the present formulation, using eqs. [6] and [7], has the *potential* for making predictions, whereas the RRKM method does not possess this capability.

<sup>2</sup>The RRKM  $k^*(E)$  functions are always lower than the empirical  $\bar{k}(E)$  functions at low energy: this is because the threshold occurs at an energy  $E_0$  which is lower than  $E_\infty$  in RRKM theory (eq. [6.5] of ref. 7), so that eq. [9] will only give the same value of  $k_\infty(T)$  at the matching temperature if the mean value of  $k^*(E)$  is below the mean value of  $k(E)$  in the significant energy range. The magnitudes of these differences viz. between  $E_0$  and  $E_\infty$ , or between  $k^*(E)$  and  $k(E)$ , become more marked the greater is the required extent of the frequency adjustment in the RRKM entropy matching procedure (cf. eq. [6.4] of ref. 7): thus, if there is a large frequency adjustment (as in Figs. 4, 6, and 10),  $k^*(E)$  will always be below  $k(E)$  in the useful energy range, but if there is only a small frequency adjustment (as in Figs. 2, 7, 8, and 9)  $k^*(E)$  will cross  $k(E)$  at high energy because it has a slightly smaller curvature, corresponding to the fact that the RRKM  $k_\infty$  increases with increasing temperature more quickly than strict Arrhenius (12).

### Cyclopropane

A property of eq. [6a] is that if the critical energy and critical frequency  $\nu_r$  are held constant, but the complexity of the molecule is increased, the ratio of active to passive states decreases somewhat; thus, it is clear that if the critical energy is considered to be localised in only one oscillator, one will not compute frequency factors much in excess of  $2 \times 10^{14} \text{ s}^{-1}$ , the value obtained for methyl isocyanide using the asymmetric C—H stretch  $\nu_1 = 2966 \text{ cm}^{-1}$  for  $\nu_r$ , as in Fig. 3a. However, if the critical energy does not have to be localised in one degree of freedom but can be distributed in some other way then, as Polanyi and Wigner recognised, frequency factors greater than  $10^{15} \text{ s}^{-1}$  are possible, via eq. [2]; our quantum reformulation of this concept is embodied in eqs. [6b] and [7b], and we will now demonstrate their application to the cyclopropane isomerisation reaction.

According to calculations by Jug (20), the reaction coordinate in the isomerisation of cyclopropane to propylene can be visualised as a superposition of the torsional vibration  $\nu_3$  and an asymmetric stretch  $\nu_{6b}$ , where these are the normal modes of vibration of a simplified model of cyclopropane, i.e.  $X_3Y_3$  and the notation is that of Herzberg (21). Extending, and being guided by, this idea, we show in Fig. 4 the results of a series of calculations for cyclopropane. We chose two normal vibrations of cyclopropane,  $\nu_3 = 1188 \text{ cm}^{-1}$ , the ring stretch, and  $\nu_6 = 3103 \text{ cm}^{-1}$ , an asymmetric  $\text{CH}_2$  stretch (19) to represent the critical motion in this reaction. The lowest two synthetic curves show that localisation of the critical energy in either one of these oscillators cannot generate the high observed reaction rate for this molecule. However, if we allow the critical energy to be shared in any fashion between these two oscillators of frequency 1188 and  $3103 \text{ cm}^{-1}$ , and accept Polanyi and Wigner's suggestion (1) that the lifetime for this complex oscillator is simply the lifetime of the oscillator having the higher frequency, then agreement with the empirical  $\bar{k}(E)$  is remarkable; also, the computed frequency factor of  $1.4 \times 10^{15} \text{ s}^{-1}$  is in excellent agreement with experiment.

Although, as noted above, the present procedure provides us with the potential for exploring different "reaction co-ordinates" in a way which is not possible using the RRKM procedure, our choice in Fig. 4 for the reaction co-ordinate for cyclopropane is not unique. We notice that Jug's picture (20) of the reaction process also includes some rotational motion of one of the  $\text{CH}_2$  groups: thus if we had chosen to share the critical energy between the ring stretch  $\nu_3$  at  $1188 \text{ cm}^{-1}$  with one or other of the

degenerate pair of  $\text{CH}_2$  twists, also at  $1188 \text{ cm}^{-1}$  (19), we would have calculated a synthetic  $\bar{k}(E)$  function only slightly below our first choice, as shown in Fig. 4, and with a frequency factor<sup>3</sup> of  $1.6 \times 10^{15} \text{ s}^{-1}$ . However, were we to share the energy equally among all three oscillators of frequency  $1188 \text{ cm}^{-1}$ , the calculated  $\bar{k}(E)$  function would be about an order of magnitude larger than the result of sharing between two.

### Non-adiabatic Reactions

The use of eq. [5] to give the rate of disruption of an oscillator involves the assumption that the process is an adiabatic one and, in general, we do not know whether this is or is not the case. The dissociation of  $\text{CO}_2$  into  $\text{CO} + \text{O}$  is, however, clearly a non-adiabatic reaction and so it is interesting at this juncture to examine the application of eqs. [6] and [7] to this reaction. The molecule has two stretching modes, a symmetric one  $\nu_1$  leading to simultaneous dissociation of both C—O bonds, and an asymmetric one  $\nu_3$  which will lead to dissociation into O and CO; it also has a pair of bending modes  $\nu_2$  which, for the time being (see Appendix), are treated as being harmonic. Moreover, it appears from our recent calculations (22) that both in  $\text{N}_2\text{O}$  and  $\text{CO}_2$ , the transmission efficiency for curve crossing is about 1 in every 100 oscillations: the results presented in Fig. 5 assume therefore that  $\nu_3$  is the critical oscillator ( $\nu_r = \nu_3 = 2349 \text{ cm}^{-1}$ ) and that a transmission coefficient  $\kappa = 0.01$  is included in eq. [5] to take rough account of the non-adiabatic problem. The agreement is reasonable, as also it is for  $\text{N}_2\text{O}$  using again  $\nu_3$  as the critical frequency ( $\nu_r = \nu_3 = 2283 \text{ cm}^{-1}$ ) and the same value of 0.01 for the transmission coefficient.

For the remainder of the paper, we will assume that the reactions we consider are adiabatic processes, although in general we have no way of knowing whether or not this is strictly true: thus, it should be kept in mind that we are always in danger of overestimating the rate constant through the neglect of a non-adiabatic transmission coefficient  $\kappa$  in eqs. [5]–[7].

### Other Molecules

In this section, we will inspect briefly the extent to which eqs. [6] and [7] are successful in reproducing the specific rate functions and observed rate constants for the usual selection of other test-case thermal unimolecular reactions.

#### Cyclobutane

We proceed here by analogy with the treatment

<sup>3</sup>Although the lower  $\bar{k}(E)$  function gives the higher  $A$ -factor, there is no inconsistency since it also gives a slightly higher activation energy, see legend to Fig. 4.

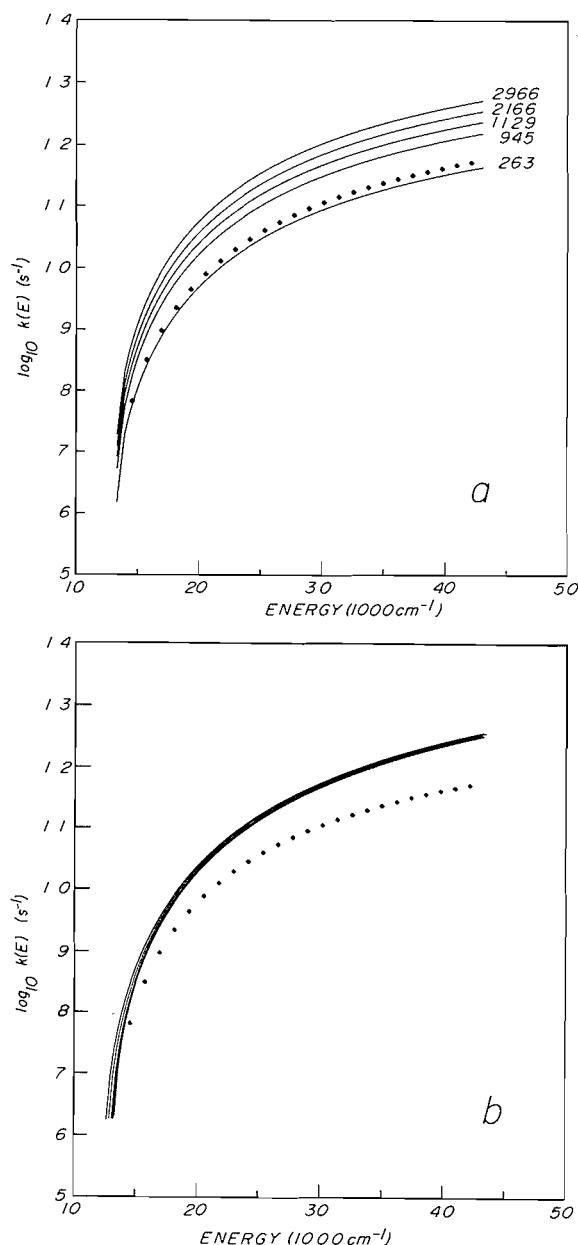


FIG. 3. (a) Comparison of five synthetic rate functions calculated according to eq. [7a] for methyl isocyanide, in the harmonic-oscillator approximation; the lines are labelled by the value of the frequency, in  $\text{cm}^{-1}$ , chosen for  $\nu_r$ . (b) Five RRKM  $k^*(E)$  functions using, in turn, the same five frequencies for the reaction co-ordinate. On both diagrams, the individual points represent the empirical  $\bar{k}(E)$  function calculated according to eq. [8], also in the harmonic-oscillator approximation.

of cyclopropane we gave above: the concentration of the critical energy in any single skeletal motion (having frequencies in the range  $200\text{--}1000\text{ cm}^{-1}$ ) gives rather low rates, and so we assume that the critical energy needs to be shared between two

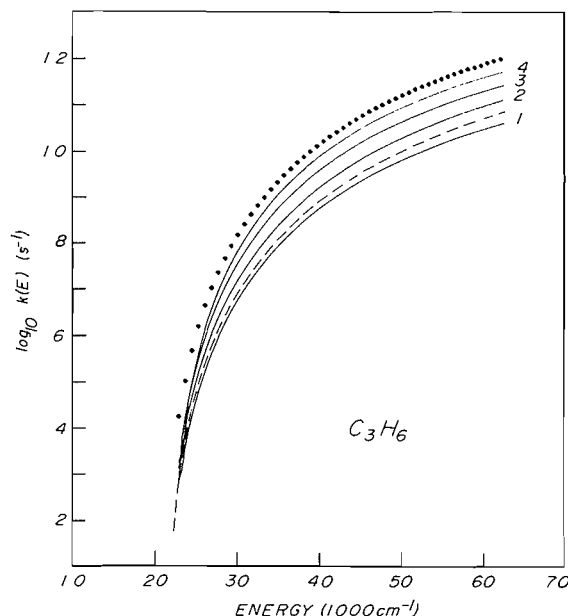


FIG. 4. Comparison of four synthetic  $k(E)$  functions (solid lines) with the empirical  $\bar{k}(E)$  function (points) and the RRKM  $k^*(E)$  function (dashed line) for cyclopropane. Curves 1 and 2 use eq. [7a] with  $\nu_r = 1188$  and  $3103\text{ cm}^{-1}$  respectively; curves 3 and 4 use eq. [7b] and share  $1188$  ( $g = 1$ ) with  $1188$  ( $g = 2$ ) and  $1188$  ( $g = 1$ ) with  $3103$  ( $g = 1$ )  $\text{cm}^{-1}$  respectively. The rate constants calculated from the synthetic functions using eq. [9] give the following Arrhenius parameters (in order 1, 2, 3, 4):  $E_\infty = 65.5, 65.5, 67.2, 66.0$  ( $65.6$ )  $\text{kcal mol}^{-1}$ ,  $A_\infty = 0.07, 0.19, 1.6, 1.4$  ( $1.9$ )  $\times 10^{15}\text{ s}^{-1}$  respectively, at  $700\text{ K}$ .

oscillational modes. Inspection of the normal modes of vibration of cyclobutane (23) shows that there is a pair having  $B_{1g}$  symmetry which form a natural choice for the reaction co-ordinate: one is the ring deformation  $\nu_9$  at  $926\text{ cm}^{-1}$  in which the  $\text{C}_1\text{--C}_2$  and  $\text{C}_3\text{--C}_4$  lengths extend simultaneously while the  $\text{C}_2\text{--C}_3$  and  $\text{C}_4\text{--C}_1$  lengths decrease; the other is the  $\text{CH}_2$  wagging  $\nu_8$  at  $1219\text{ cm}^{-1}$  which simultaneously moves the  $\text{H}_2$  pairs towards (and away from) the line of the  $\text{C--C}$  bonds, as required by the end of the dissociation process. Figure 6 compares the computed  $k(E)$  function based on this assumption with the empirical  $\bar{k}(E)$  and RRKM  $k^*(E)$  functions: the computed rate constants are about a factor of 3.5 too high when compared with the experimental values between  $700$  and  $800\text{ K}$ .

#### Ethyl Chloride

Figure 7 shows the results of choosing the critical coordinate to be one of the  $\text{C--C--Cl}$  deformation  $\nu_{11} = 336\text{ cm}^{-1}$ , the  $\text{C--Cl}$  stretch  $\nu_{10} = 671\text{ cm}^{-1}$  or the  $\text{CH}_3$  degenerate stretch  $\nu_2 = 2946\text{ cm}^{-1}$  (19). The choice of the  $\text{C--Cl}$  stretch  $\nu_{10}$  gives almost exact agreement for the  $k(E)$  function with the calculated rate constant some  $20\text{--}25\%$  below the ob-

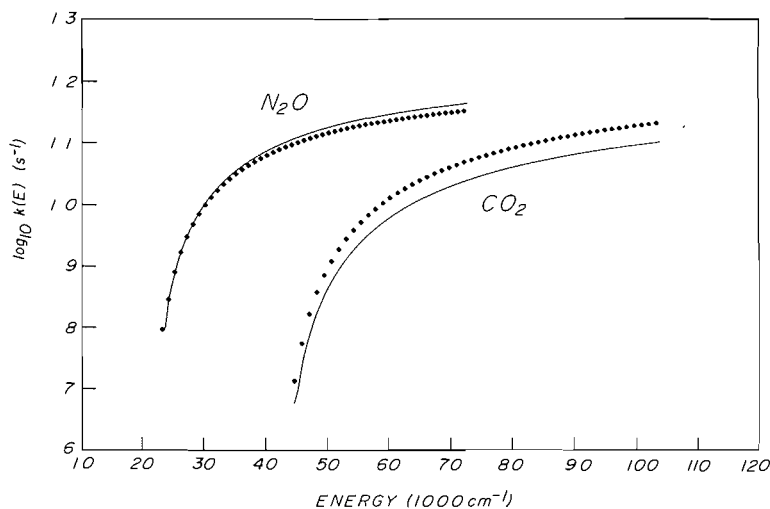


FIG. 5. Comparison of synthetic and empirical rate functions for carbon dioxide and nitrous oxide. The rate constants calculated from the synthetic functions using eq. [9] give the following Arrhenius parameters:  $\text{CO}_2$ ,  $E_\infty = 124$  (127)  $\text{kcal mol}^{-1}$ ,  $A_\infty = 1.2$  (2.0)  $\times 10^{12} \text{ s}^{-1}$  at 3000 K;  $\text{N}_2\text{O}$ ,  $E_\infty = 66$  (66)  $\text{kcal mol}^{-1}$ ,  $A_\infty = 1.9$  (1.5)  $\times 10^{12} \text{ s}^{-1}$  at 2000 K.

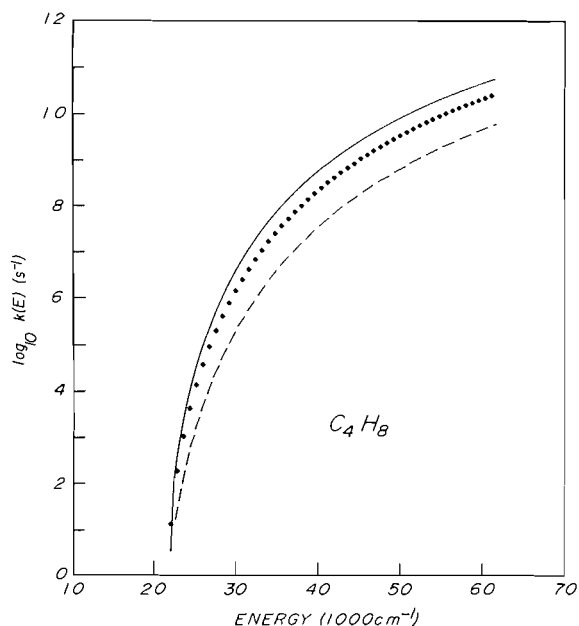


FIG. 6. Comparison of synthetic, empirical, and RRKM rate functions for cyclobutane. The rate constants calculated from the synthetic rate function according to eq. [9] give the following Arrhenius parameters:  $E_\infty = 62.9$  (62.5)  $\text{kcal mol}^{-1}$ ,  $A_\infty = 16$  (3.7)  $\times 10^{14} \text{ s}^{-1}$  at 700 K.

served values. In this calculation, the synthetic  $k(E)$  curves do not fall exactly into a sequence as happens in Fig. 3a: this is because  $\nu_{11} = 336 \text{ cm}^{-1}$  is treated as being harmonic, whereas  $\nu_{10} = 676$  is assigned an anharmonicity  $x_e = 0.006$  and  $\nu_2 = 2946 \text{ cm}^{-1}$  has  $x_e = 0.018$ ; in fact, were we to imagine that any one of  $\nu_4 = 1463 \text{ cm}^{-1}$  ( $\text{CH}_3$  degener-

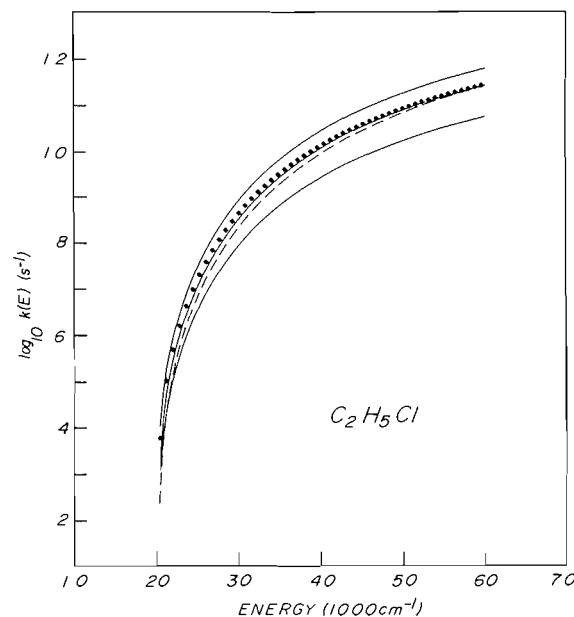


FIG. 7. Comparison of three synthetic rate functions, calculated according to eq. [7a] with the empirical and RRKM rate functions for ethyl chloride: the synthetic functions, from top to bottom, use  $\nu_7 = 2946$ , 676, and  $336 \text{ cm}^{-1}$  respectively, and using eq. [9], give rise to the following Arrhenius parameters:  $E_\infty = 58.4$ , 58.6, 58.4 (58.4)  $\text{kcal mol}^{-1}$ ,  $A_\infty = 2.0$ , 0.9, 0.2 (1.1)  $\times 10^{14} \text{ s}^{-1}$  at 600 K.

ate deformation) or  $\nu_5 = 1448 \text{ cm}^{-1}$  (scissors) or  $\nu_{14} = 1448 \text{ cm}^{-1}$  ( $\text{CH}_3$  degenerate deformation) (19), all treated as harmonic in our empirical assignment of anharmonicity, was the critical oscillator, the computed  $k(E)$  function would be indistinguishable on

the scale of the diagram from the one shown for  $\nu_{10} = 676 \text{ cm}^{-1}$ ,  $x_e = 0.006$ . Thus, although Fig. 1 shows that the deduction of the Slater-Forst  $k(E)$  function is relatively insensitive to the anharmonicities chosen for the various oscillators in the molecule, our computed  $k(E)$  function is sensitive to the anharmonicity chosen for the critical oscillator itself; the reason for this is quite clear on inspection of eq. [6a], since the spacings of the  $\epsilon_r(n_r)$  depend strongly upon the value of  $x_e$  in the reactive region.

#### Ethyl Isocyanide

Ethyl isocyanide possesses two bending frequencies, cf., ethyl cyanide (19),  $\nu_{13} = 226 \text{ cm}^{-1}$  and  $\nu_{20} = 270 \text{ cm}^{-1}$  (24). Figure 8 shows the empirical  $k(E)$  function, the RRKM  $k^*(E)$  function and two synthetic  $k(E)$  functions, one for a single oscillator ( $\nu_r = \nu_{13} = 226 \text{ cm}^{-1}$ ) and the other for which the critical energy is shared between the two bending motions: the latter is rather better.

#### Trideuteriomethyl Isocyanide

Figure 9 shows the results for  $\text{CD}_3\text{NC}$  analogous to the calculations presented for  $\text{CH}_3\text{NC}$  in Fig. 2: considering the excellent agreement with experiment for  $\text{CH}_3\text{NC}$ , the results shown in Fig. 9 for

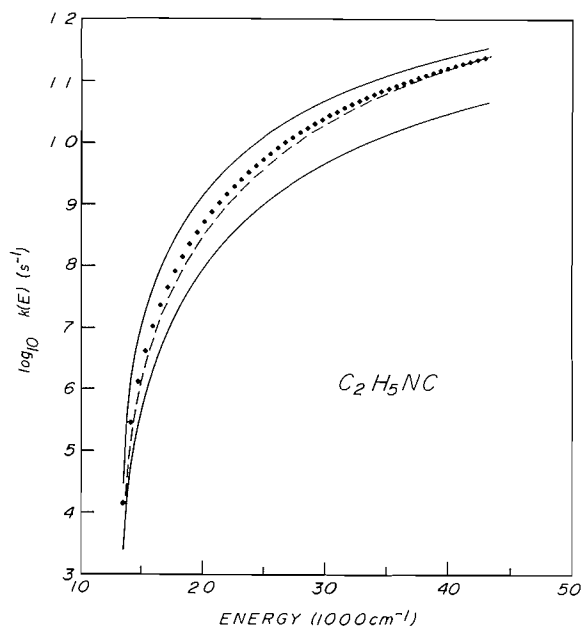


FIG. 8. Comparison of two synthetic rate functions with the empirical and RRKM rate functions for ethyl isocyanide: the lower synthetic function uses eq. [7a] and one bending frequency, the upper one uses eq. [7b] and both bending frequencies. The rate constants calculated from these synthetic rate functions according to eq. [9] give the following Arrhenius parameters:  $E_\infty = 38.3, 38.1 (38.4) \text{ kcal mol}^{-1}$ ,  $A_\infty = 1.5, 31 (7.8) \times 10^{13} \text{ s}^{-1}$  at 500 K.

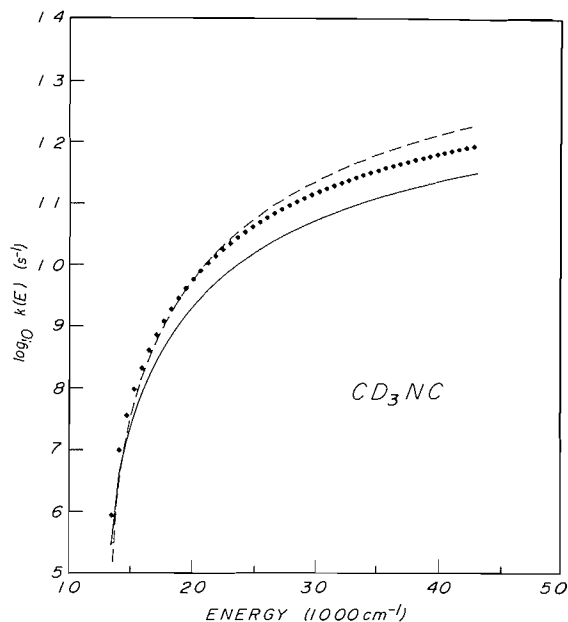


FIG. 9. Comparison of synthetic, empirical, and RRKM rate functions for trideuteriomethyl isocyanide. The rate constants calculated from the synthetic rate function according to eq. [9] give the following Arrhenius parameters:  $E_\infty = 38.5 (38.5) \text{ kcal mol}^{-1}$ ,  $A_\infty = 1.7, (4.5) \times 10^{13} \text{ s}^{-1}$  at 500 K.

$\text{CD}_3\text{NC}$  are somewhat disappointing, but nevertheless the predicted rate is well within a factor of 3 of the correct result.

#### Ethane

The ethane molecule possesses 6 singly degenerate and 6 doubly degenerate normal modes of vibration (ref. 21, p. 115). The ethane decomposition also has a very high frequency factor (26) of about  $6.3 \times 10^{16} \text{ s}^{-1}$ , so it is clear that  $m > 1$  in eq. [6b]. Examination of the normal-mode diagrams shows that there are only three of them which lead to any significant extension and contraction of the C—C bond, namely  $\nu_1 = 2954 \text{ cm}^{-1}$  ( $\text{CH}_3$  symmetric stretch),  $\nu_2 = 1388 \text{ cm}^{-1}$  ( $\text{CH}_3$  symmetric deformation), and  $\nu_3 = 995 \text{ cm}^{-1}$  (C—C stretch) (19). Sharing of the critical energy of reaction between these three degrees of freedom yields a synthetic  $k(E)$  function remarkably close to the empirical one, with the rate constant only about a factor of 3 too low.

#### Discussion

In this paper, we have taken the original concept of Polanyi and Wigner and reformulated it in discrete or quantised form.<sup>4</sup> As a result, we have

<sup>4</sup>As such, this reformulation is probably the nearest attainable approach to the quantum (anharmonic) form of the Slater theory, which so eluded Noel Slater, and others, over the years.

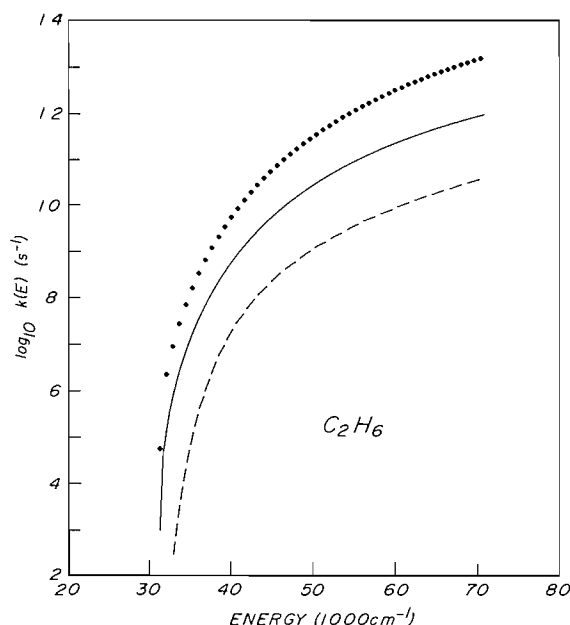


FIG. 10. Comparison of synthetic, empirical, and RRKM rate functions for ethane. The rate constants calculated from the synthetic rate function according to eq. [9] give the following Arrhenius parameters:  $89.8$  ( $89.2$ )  $\text{kcal mol}^{-1}$ ,  $A_{\infty} = 2.7$  ( $6.3$ )  $\times 10^{16} \text{ s}^{-1}$  at  $900 \text{ K}$ .

found that the rate of the unimolecular disruption of a rotationally and vibrationally excited molecule can be calculated with reasonable accuracy (usually within a factor of 3 or 4) by a very simple procedure. One first identifies the type of vibrational motion which, if carried to its limit, will result in the required reaction process. For simple molecules, it appears that this is always a simple process, and can be represented as the concentration of the critical energy in a *single* oscillator; for more complicated molecules it appears that, in general, the critical energy can be shared among two or perhaps three oscillators, and the choice of such oscillators is not very difficult in certain cases, e.g. cyclobutane, or possibly ethane. Then one enumerates the states of the molecule which have more than the critical energy localised in the critical oscillator(s) and assigns to them a mean lifetime which is simply half the vibrational period of the fastest critical oscillator: the remainder of the states are treated as being passive, and the synthetic  $k(E)$  function is just the product of the mean decay rate constant for an active state with the ratio of the number of active to passive states.

It would be wise not to read too much physical significance into the apparent success of this simple model at this early stage, for several reasons. First, each state is treated as being discrete and infinitely narrow, whereas in reality each level is broadened

from a number of causes, and in fact each level is overlapped by many of its neighbours: we intend to address this problem in a later publication.

Second, we have used a normal-mode representation of the internal motions of the molecule, whereas at these moderately high levels of excitation, a local-mode representation may be more appropriate: we are assuming here, of course, that the normal-mode and local-mode representations are commensurable in the sense that either will give us about the same densities of passive and active states in the given energy region. Hence, the mapping of the normal modes into the local modes would be a prerequisite for the proper choice of the critical oscillator for each reaction. It is clear, however, that the actual disruption process is often going to be representable by the concentration of the critical energy into a single local mode: as examples, we have in this paper the cases of methyl isocyanide and ethyl chloride: the latter result is rather surprising in view of the amount of effort which has been expended in the past trying to formulate the nature of the transition state<sup>5</sup> in such four-centre reactions! For such cases, the principal requirement is a knowledge of the frequency of the local mode leading to reaction<sup>6</sup>. For more complicated molecules, the critical energy can be shared between two, or perhaps even three vibrational degrees of freedom, and by using (not illogical) combinations of such oscillations, it is fairly easy to generate acceptable  $k(E)$  functions and rate constants: this sharing of the critical energy among two or more oscillators leads directly to the prediction of "high" frequency factors (1, 8, 9, 27).

Third, our knowledge of the energy-level density in any molecule must be very imperfect because, in general, we have very scant knowledge of anharmonicities and of rotation-vibration interaction constants, both of which affect the level densities

<sup>5</sup>We always have to keep in mind, of course, that a curve-crossing may be involved, and that we are really dealing with a more complex (and therefore faster) process moderated by a non-adiabatic transmission factor  $\kappa$ .

<sup>6</sup>In this context, the apparent discrepancy noted above between the success of the method for methyl isocyanide and for its isotopic variant becomes explicable. In methyl isocyanide, we have assumed equally spaced harmonic levels of the  $263 \text{ cm}^{-1}$  oscillation in evaluating eq. [6a], whereas, above threshold, this oscillation must degenerate into a rotation with increasing level spacings as the energy increases. Likewise, in the deuterated molecule, the analogous bending at  $249 \text{ cm}^{-1}$  also degenerates into a rotation, but because of the increased effective moment of inertia for this rotation, the level spacings will be closer than for the normal molecule. Thus, in the final analysis, the  $k(E)$  function for the heavier molecule will not be a factor of three below that for the normal molecule, as we find here using our normal-mode approximations; similar considerations might apply also to ethyl isocyanide.

significantly. We have tried to allow for the former in a very empirical manner, but have ignored completely the problem of rotation-vibration interaction, although we have allowed for the density of rotational states at all stages in the formulation: this is analogous to the approximations we have made previously in treating the dissociations of CO<sub>2</sub> and N<sub>2</sub>O, both at high (22) and low (4) pressures, but it is clear, especially in the former case, that the neglect of rotation-vibration interaction has an adverse effect on the quality of the computed rate constants; there is, of course, no *a priori* reason why one should not consider the critical energy to be shared between rotation and vibration, but we have not explored the behaviour of such calculations.

In conclusion, we believe that the present approach has a number of favourable qualities. It is simple in concept, and simple to use: moreover, for simple molecules (fewer than 5 or 6 atoms, say) there usually appears to be a fairly unambiguous choice of critical oscillator which causes eqs. [6] and [7] to give results within a factor of about 3 of the empirical Slater-Forst  $k(E)$  function. For such molecules, it is now becoming possible to compute, *ab initio*, potential surfaces of "chemical accuracy" from which all the required molecular parameters, i.e. potential barrier, vibration frequencies, moments of inertia, and so on can be derived: thus, a true *ab initio* calculation of a unimolecular rate constant becomes a possibility. For more complicated molecules, this approach gives hope that we will be able eventually to distinguish between alternative pathways in a manner which is not possible with the RRKM method: this is because we do not have to adjust the frequencies of the molecule to match the observed frequency factor, but rather we take the natural frequencies as best we can determine them, and by the appropriate choice of a reaction co-ordinate are led to the unambiguous calculation of the frequency factor. Thus, this approach has considerably greater predictive potential when one is interested in the rates of reactions for which no measurements are possible at infinite pressure; however, before true predictions can be made for reactions in general, a period of exploration is required in which we examine, for a series of reaction types, the correspondence between the structure of the molecule and the nature of the appropriate reaction co-ordinate.

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## Appendix

### Anharmonicities and Other Molecular Constants

In general, the calculations reported in this paper use the same set of molecular and reaction rate constants as we used previously, viz. Tables 2 and 3 of ref. 12; ethane was not considered in that paper, but the vibration frequencies (19), moments of inertia (6), the barrier to internal rotation (6) and the rate parameters (26) are taken from recent reliable sources. However, the anharmonicities we used previously are rather unsatisfactory for the present application. In that work (12) we examined the variation in density of states for methyl isocyanide and for cyclopropane as a function of the anharmonicity chosen for certain oscillators, and from the extent of that variation, it became clear that the choice of anharmonicities would have only a moderate effect on the derived  $k(E)$  function,

essentially since eq. [8] contains a ratio of terms which are changed in the same sense by changes in anharmonicity. Based on Cyvin's treatment of cyclopropane (14), we had recommended some empirical anharmonicity factors for use in these calculations (12) but these anharmonicities are, in general, too large: for example, the bending frequency of  $263\text{ cm}^{-1}$  in methyl isocyanide would "dissociate" before the critical energy of  $13\,361\text{ cm}^{-1}$  is reached, making the application of eq. [6a] impossible. These large anharmonicities, although acceptable for general statistical calculations, are unreasonable for molecular-dynamic applications. Consequently, we have adopted an empirical method of assigning anharmonicity factors based upon the nature of the vibration in question and its estimated dissociation limit. We assign an effective dissociation limit  $D_i$  to each vibration: for example, the C—C stretch in ethane would have  $D_i \approx 90\text{ kcal mol}^{-1}$ , the symmetric stretch of a  $\text{CH}_3$

group would have  $D_i \approx 300\text{ kcal mol}^{-1}$  and the asymmetric stretch of the same group about  $100\text{ kcal mol}^{-1}$ ; all bending or deformation degrees of freedom, which cannot be assigned a dissociation energy in this simple way, were assumed to be harmonic. To simplify the procedure even more, it matters very little what effective dissociation energy  $D_i$  is assigned to an oscillator once  $D_i > 2E_{\text{crit}}$ ; consequently, all anharmonic oscillators were given a number  $N_i$  in the range  $1 \leq N_i \leq 2$  where  $N_i = D_i/E_{\text{crit}}$  for  $D_i \leq 2E_{\text{crit}}$  or  $N_i = 2$  for  $D_i > 2E_{\text{crit}}$ , and the anharmonicity factor was then calculated according to the standard Morse-oscillator equation (25).

$$[\text{A1}] \quad \omega_e x_e = \omega_e^2 / 4N_i E_{\text{crit}}$$

The only exception to the use of this recipe was in the case of  $\text{N}_2\text{O}$ , where reliable values for the anharmonicities exist, cf. Table 1 of ref. 4.